Allegrow Biotech

AimGel

Artificial Cells for Immune Cell Activation

Biomimetic

Modular

Feeder-free

T cells & NK cells

Human, murine & canine

Website
www.allegrowbiotech.com

Email
welovecells@allegrowbiotech.com

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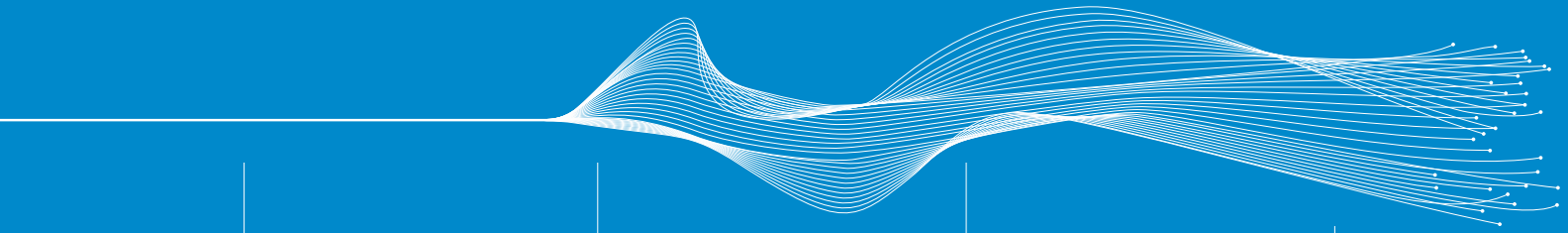
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About Us

Allegrow Biotech™ is a biotech specialising in immunomodulation via biomimetic approaches, licensing technology from the Hong Kong University of Science and Technology. The core team is based in Hong Kong at The Hong Kong Science and Technology Park, with presence spread across China mainland, United Kingdom and the U.S. We endeavour to open a new chapter in cell signalling and modulation with our AimGel™ technology.



2022

Allegrow Biotech founded

2024

Established research HQ at Hong Kong Science & Technology Park

2025

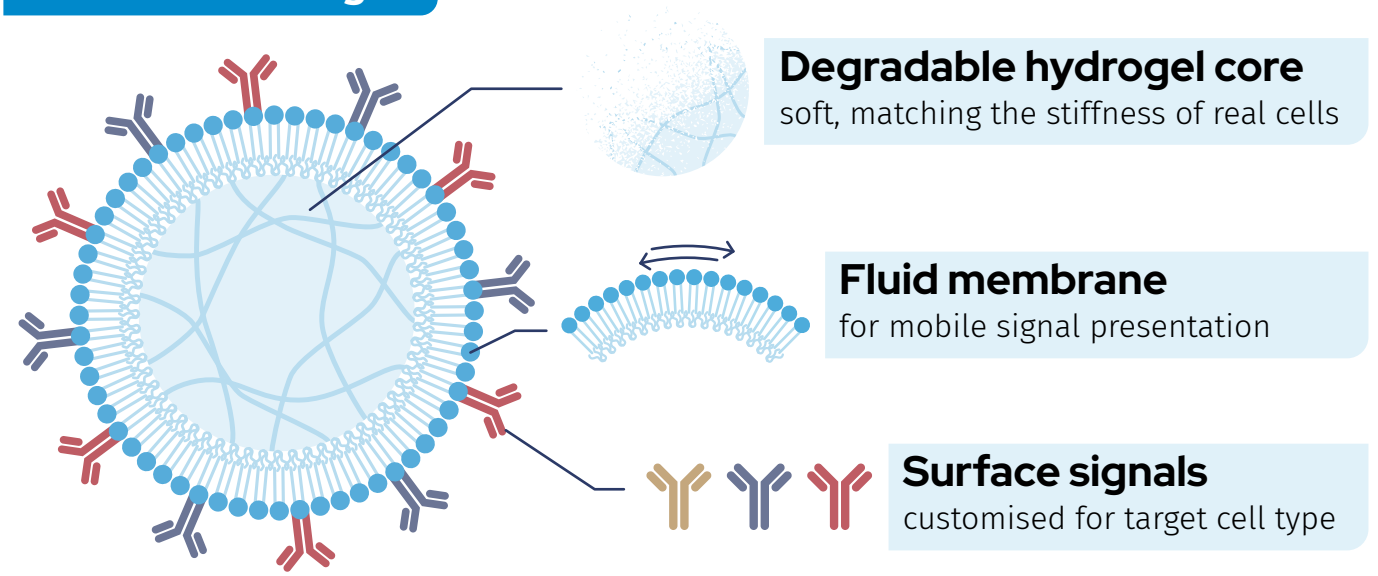
AimGel launch:
T & NK cells activators

2026

Expand AimGel lineup to mouse, dog & AimCore

AimGel A modular artificial cell platform

Biomimetic design



Degradable hydrogel core
soft, matching the stiffness of real cells

Fluid membrane
for mobile signal presentation

Surface signals
customised for target cell type

AimGel™ artificial cells are designed from the ground up to mimic natural antigen presenting cells (APCs). The soft hydrogel core replicates the mechanical stiffness and size (7-12 µm) of real cells, while activation signals move freely across the surface, just as they would on a natural APC membrane.

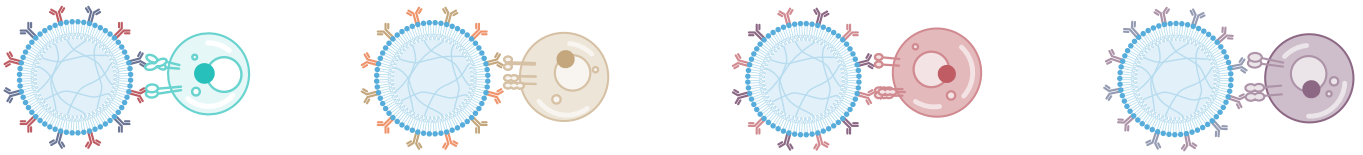
This biomimetic design delivers measurable advantages: AimGel requires significantly fewer activation signals than rigid magnetic microplastic beads while achieving more effective yet gentler activation. The result is higher cell expansion, better viability, and reduced exhaustion, cells remain responsive and can be re-stimulated.

One platform, unlimited applications

Degradation kinetics, microparticle size, and signal loading capacity are all chemically tuneable, giving you precise control over activation parameters. Load different signals for different experimental needs, currently optimised for T cell (human, murine, canine) and NK cell (human) activation, with other cell types (Treg) and applications (subtype biasing) coming soon.

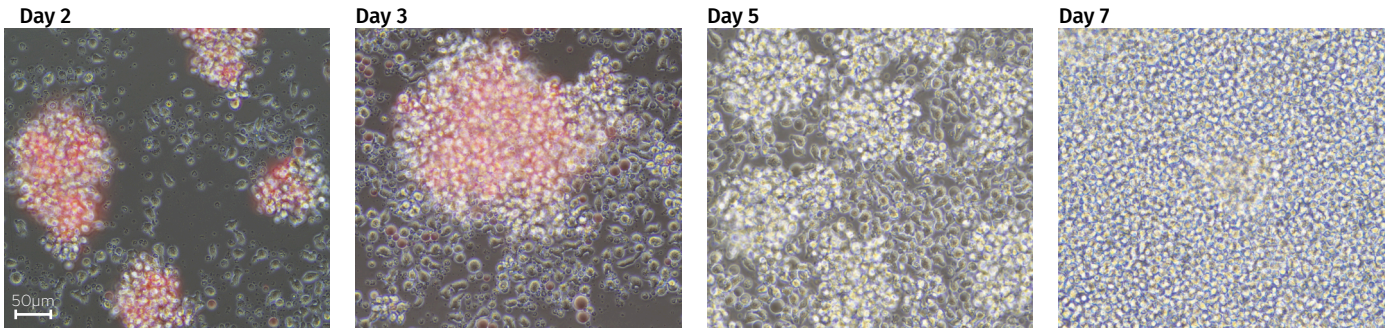
The platform adapts to your research: adjust activation duration, coordinate with lentiviral transduction timing, or engineer entirely new signal combinations using our AimCore™ lineup of blank microparticles.

Our AimGel activators ship non-hydrolysable and are cleared with the included lysis buffer. If your workflow needs them to clear on their own at a set point, we can formulate them to hydrolyse under your culture conditions.



AimGel A modular artificial cell platform

Controlled, safe degradation



T cells form clusters with AimGel (dyed in red)

T cells-AimGel clusters grow

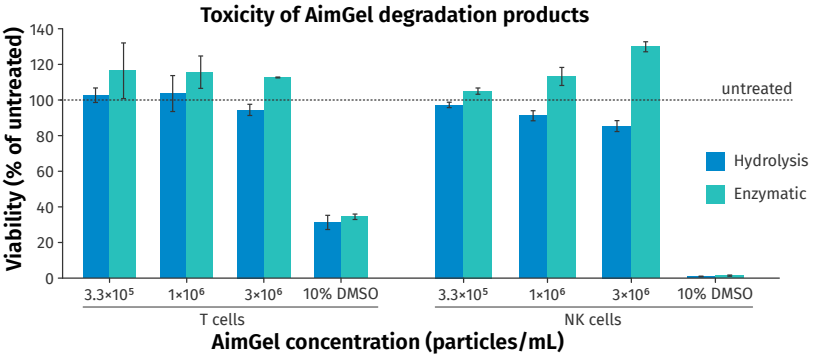
AimGel self-degrades*, T cell-AimGel clusters dissociate

AimGel fully degraded, releasing T cells

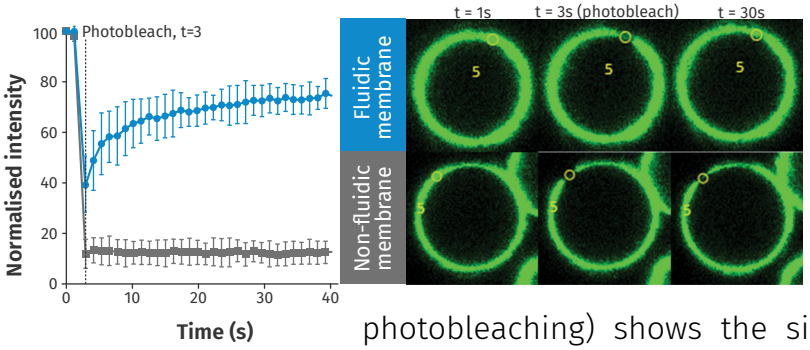
*Shown with hydrolysable AimGel; our activators ship non-hydrolysable.

Removal takes one step. Add the included lysis buffer and the microparticles are fully degraded in 30 minutes at 37 °C. Dextranase clears the hydrogel core without affecting the protein signals or your cells. Nothing is separated out. The microparticles are digested in place rather than pulled away from your culture. No magnet, no column, and no cells lost to a separation step.

The degradation products from AimGel™ microparticles, be it from their own hydrolysis or the enzymatic degradation via the lysis buffer, do not affect human T & NK cell viability, compared to 10% DMSO cytotoxic control.



Fluid, mobile membrane



On a rigid bead, signals stay where they are attached. On AimGel they diffuse freely, so surface ligands reorganise and localise as a cell engages, the way receptors form clusters at a natural immune synapse.

FRAP (fluorescence recovery after photobleaching) shows the signal recovery on a fluid membrane within seconds, while the non-fluidic control stays dark.

Defined platform

The AimGel artificial cell platform is chemically defined, from core to lipid membrane to surface modifications. AimGel activators are built on this platform by docking cell-specific signalling antibodies at defined densities and ratios, so activation is reproducible between runs.

AimGel A modular artificial cell platform

Less prep

AimGel
Mix & grow with one step enzymatic (dextranase) removal

Magnetic beads
Magnetic separation, with potential cell loss

Coated plates
Overnight coating then washing

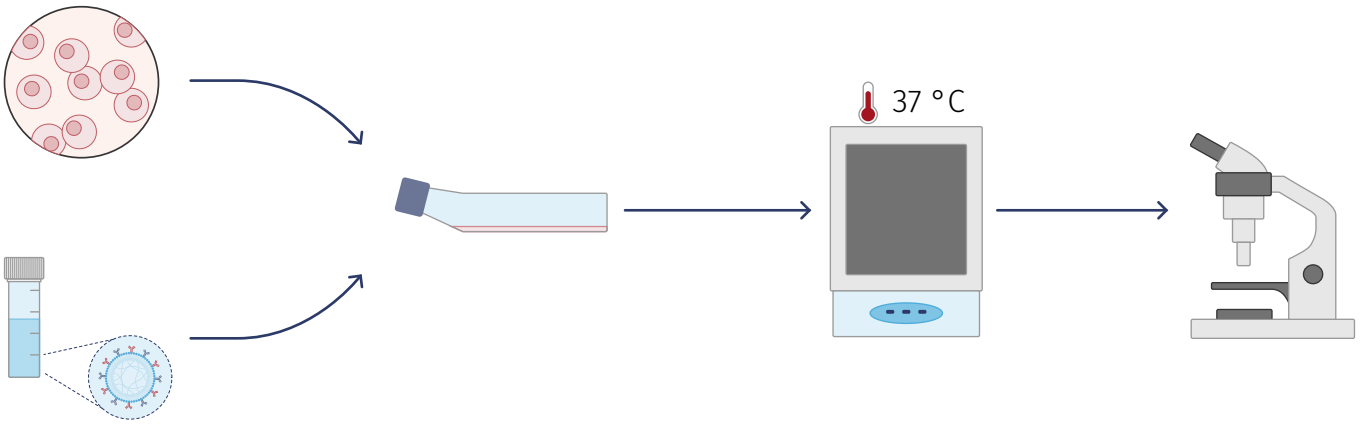
Feeder cells
Cell-line maintenance irradiation/chemical cell killing

Skip isolation steps

Whether you're starting with PBMCs or isolated cells[1], AimGel™ activates both efficiently, achieving >90% activation in your target cell population.

1: Skips isolation steps from PBMC for T cells: human, mouse & dog. Isolation steps still recommended for NK cells.

Sample workflow

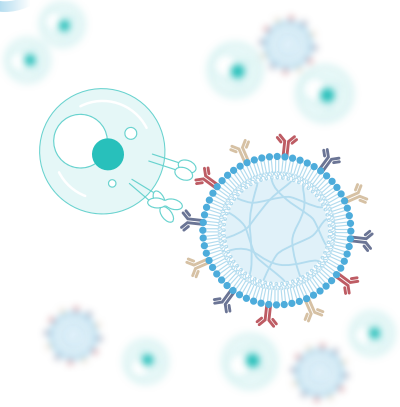


Step 1	Suspend AimGel with cells in culture vessel according to ratio
Step 2	Incubate in your incubator of choice
Observe	Cells enter log growth phase the next day

You could also re-stimulate the cells according to your needs

AimGel Human T Cell Activator

ATC01-0050H

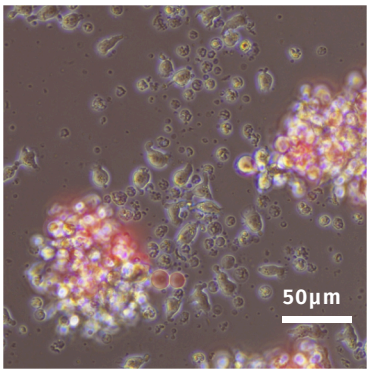


3 signals

- anti-CD3
- anti-CD28
- anti-CD137

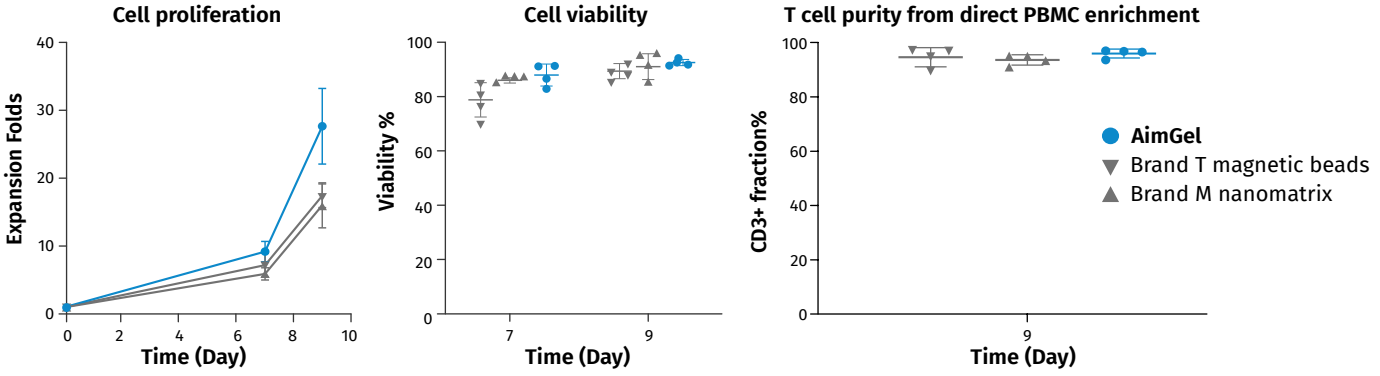
Compatible with

PBMC or isolated CD3+ T cells

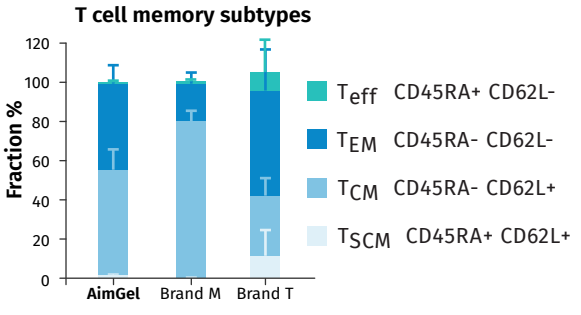


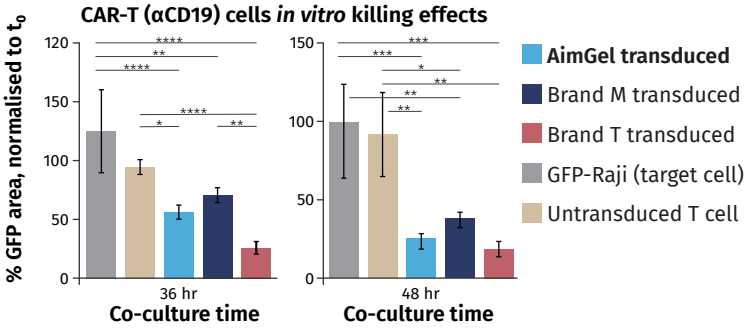
Kit contents	1 vial 0.5 mL at 4×10^7 microparticles/mL (2×10^7 total), in PBS with 0.5% P/S
stable at 4 °C for 24 months	1 vial lysis buffer: 100× dextranase in PBS

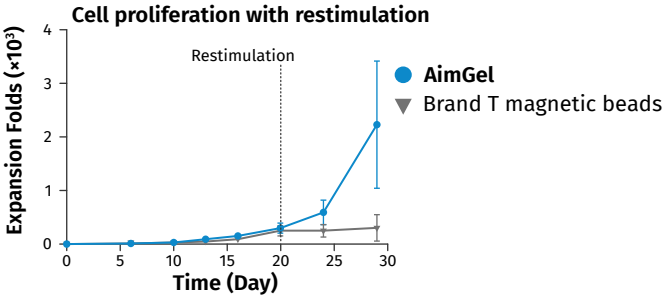
Activate and expand human T cells directly from PBMC. Get 60% more cells when expanding straight from PBMC when compared with magnetic beads and nanomatrix activators, with matching viability and purity.



AimGel™ produces more even fractions of effector memory and central memory subtypes. T cells expanded and activated by AimGel and then transduced showed matching cytotoxic effects compared with T cells expanded and activated by magnetic beads and nanomatrix.



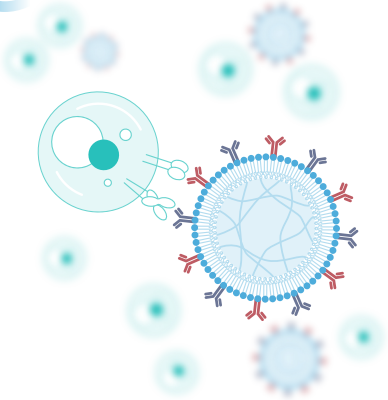




With a gentler, more biomimetic stimulation that requires fewer signals/ligands, AimGel is able to restimulate cells quicker when compared to magnetic beads.

AimGel Mouse T Cell Activator

ATC01-0050M

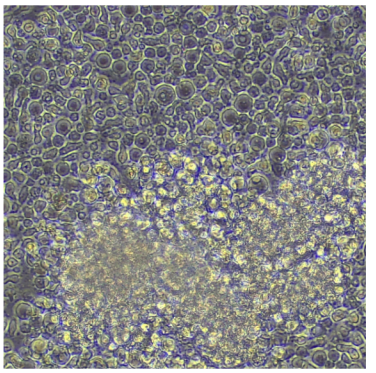


2 signals

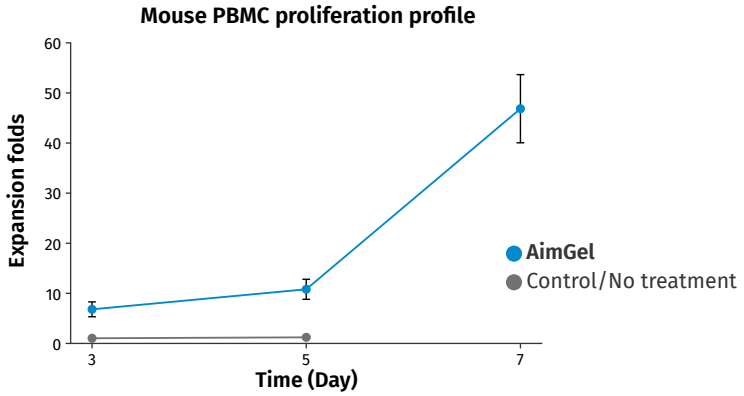
anti-mouse CD3
anti-mouse CD28

Compatible with

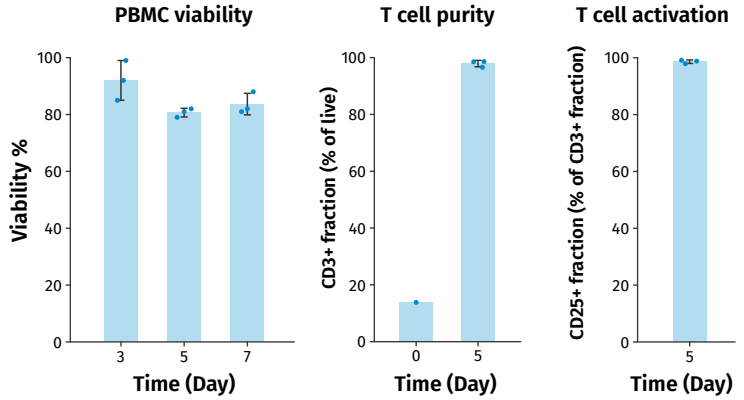
PBMC or isolated CD3+ mouse T cells



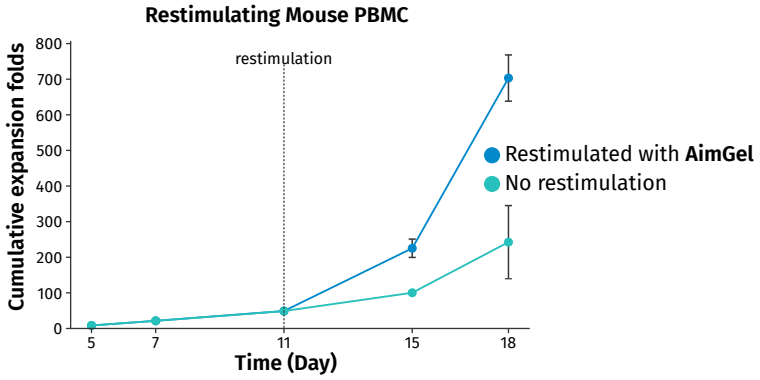
Kit contents	
stable at 4 °C for 12 months	1 vial 0.5 mL at 4×10⁷ microparticles/mL (2×10⁷ total), in PBS with 0.5% P/S
	1 vial lysis buffer: 100× dextranase in PBS



Activate and expand mouse T cells directly from PBMC, with no isolation step. AimGel™ treated PBMC reaches ~45-fold expansion by day 7.



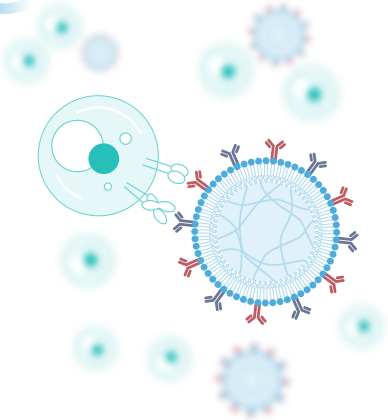
AimGel also enriches: the CD3+ fraction rises from 14% to 98% by day 5, with 98% of those cells CD25+ activated. Factoring in PBMC expansion fold, T cell fraction expanded ~330-fold (approximating day 5 fraction for day 7).



Re-stimulating at day 11 roughly triples the yield. Cumulative expansion reaches ~700-fold by day 18, against ~240-fold without re-stimulation. All cells were first activated with AimGel prior to restimulation.

AimGel Dog T Cell Activator

ATC01-0050D

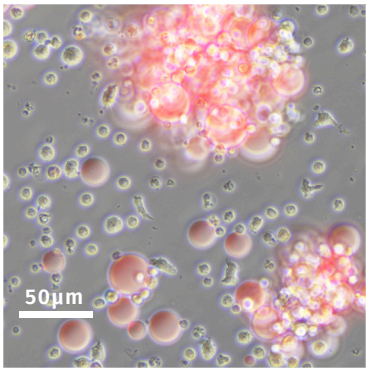


2 signals

anti-dog CD3
anti-dog CD28

Compatible with

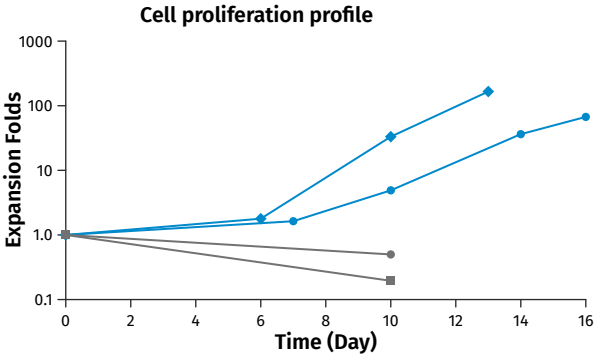
PBMC or isolated CD3+ dog T cells

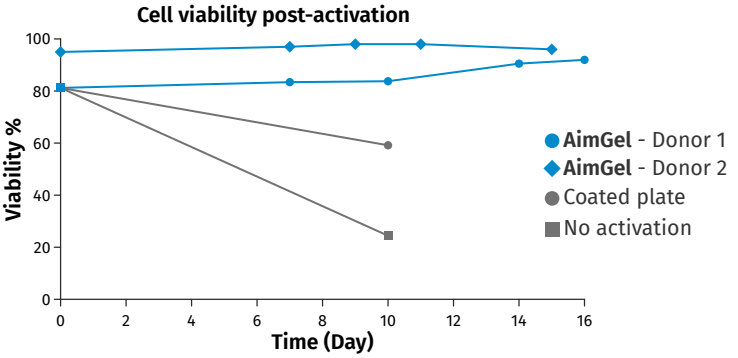


Kit contents	1 vial 0.5 mL at 4×10^7 microparticles/mL (2×10^7 total), in PBS with 0.5% P/S
stable at 4 °C for 12 months	1 vial lysis buffer: 100× dextranase in PBS

Expansion of dog PBMC with AimGel™ Dog T Cell Activator compared with well plates coated with dog anti-CD3/CD28 antibodies and negative control. By day 10, both coated-plate and unactivated cultures had fallen below their starting number. AimGel with dog anti-CD3/CD28 antibodies expanded 68-fold and 165-fold in two donors, by day 16 and day 13 respectively. Controls (coated plate and negative) done with donor 1 only.

Cells (PBMC) maintain high viability post-activation, with 2 dog donors. Compared with coated plate and no activation controls (donor 1 only).





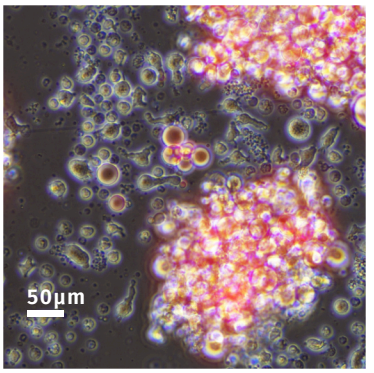
AimGel Human NK Cell Activator

ANK01-0050H

Compatible with

Isolated human CD3- CD56+ NK cells

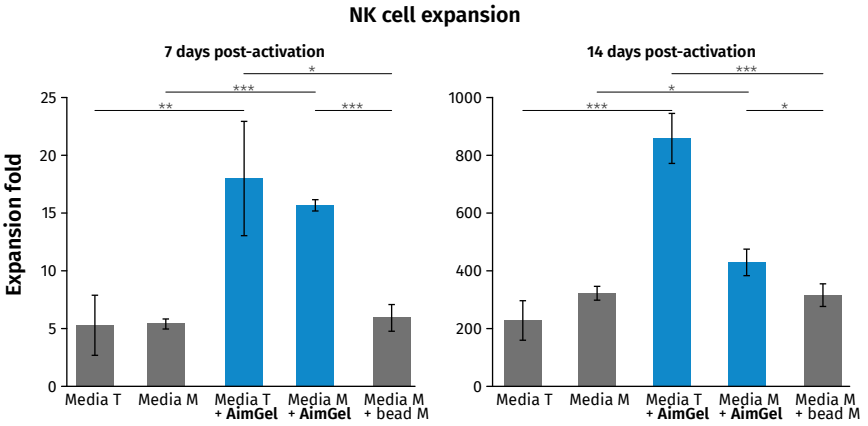
While culturing from PBMC is possible, it is dependent on the donor's starting NK fraction size.



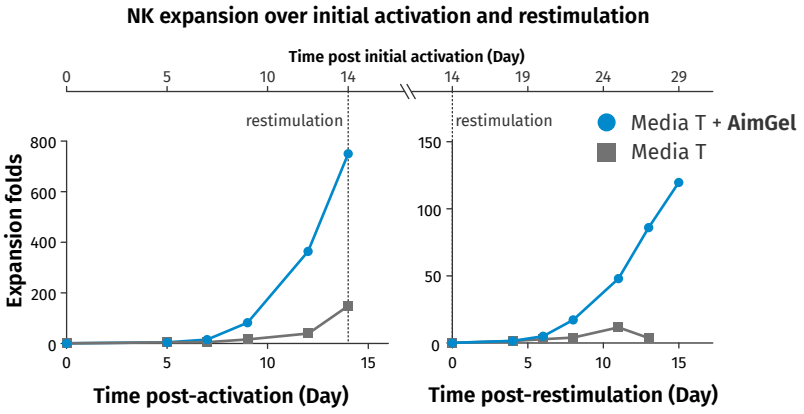
Kit contents

stable at 4 °C for 12 months

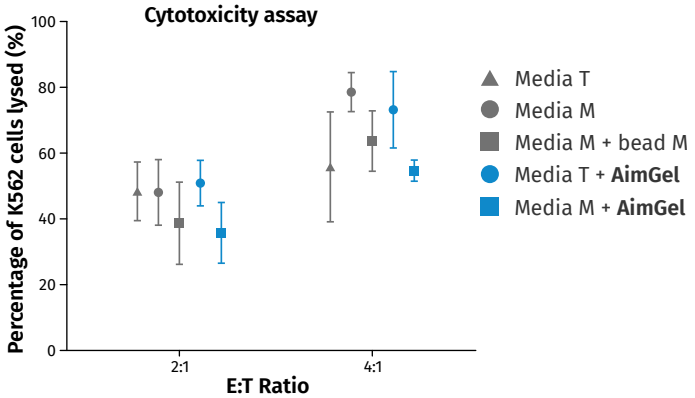
- 1 vial 0.5 mL at 4×10^7 microparticles/mL (2×10^7 total), in PBS with 0.5% P/S
- 1 vial lysis buffer: 100× dextranase in PBS



AimGel™ for human NK cells works with the NK media you already use. Added to two commercial NK media, it raised day-14 expansion from 230- to 860-fold in Media T, and from 325- to 430-fold in Media M. It also outperformed the bead activator sold alongside one of those media (M). Mean ± SD, n=4. * p<0.05, ** p<0.01, *** p<0.001.



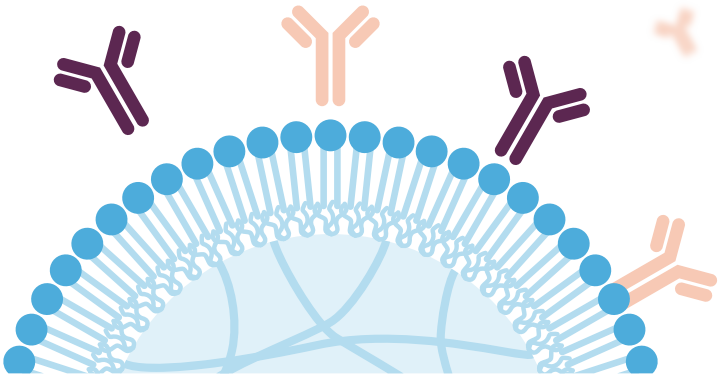
Re-stimulating at day 14 opens a second expansion phase. AimGel-activated NK cells expanded a further 120-fold over the following two weeks, where cells in medium alone peaked around 12-fold and then declined.



Cytotoxic function is unaffected. Against K562 targets at 2:1 and 4:1 effector-to-target ratios, AimGel-expanded NK cells lysed as effectively as cells grown in medium alone or with the bead kit. Function is preserved while yielding more cells.

AimCore Blank artificial cells

AVI(01 - 12)-0100



Signals
any biotinylated ligand

Compatible with
Cell types and species of your choice

Kit contents	
stable at 4 °C for 24 months	1 vial 1.0 mL at 4×10⁷ microparticles/mL, in 0.9% NaCl with 0.5% P/S
	1 vial lysis buffer: 100× dextranase in PBS

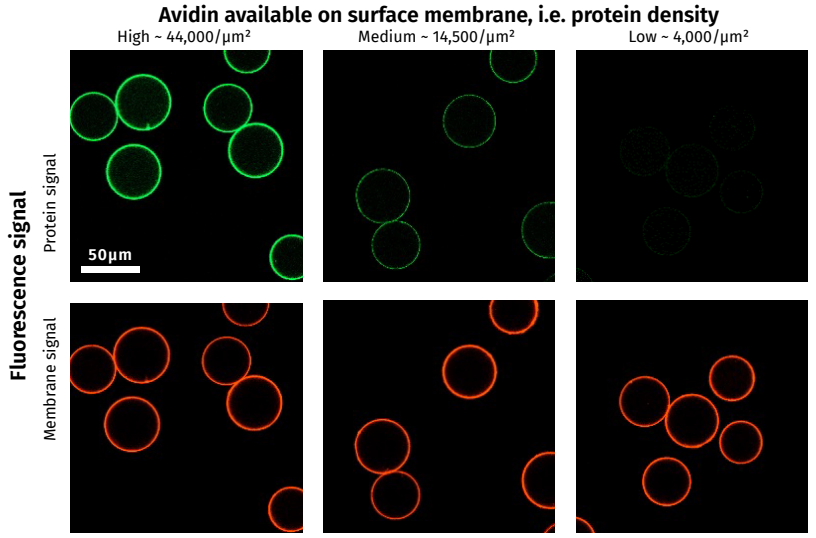
AimCore™ is a blank canvas for cell signalling, enabling you to dock any biotinylated signals onto the surface for cell-like presentation.

Biotinylated signals just attach onto AimCore's fluid surface membrane via simple mixing.

We offer a selection of AimCore microparticles to give users choices in removal methods in the culture media (degradability), signal loading capacity (avidin amount) and membrane fluidity.

All AimCore formulations are removable via enzymatic lysis (the provided dextranase), while hydrolysable AimCore formulations self-degrade in 7-10 days under physiological conditions.

Degradability	hydrolyses under physiological conditions or via enzymatic lysis only
Membrane fluidity	low or high
Avidin loading	~4,000 / ~14,500 / ~44,000 per µm²



For example, our AimGel™ Dog T Cell Activator is based on the AVI11-0100 AimCore, which is persistent (non-hydrolysable), with high membrane fluidity and a medium signal loading capacity (~14,500 molecules of avidin per µm²). See our catalogue for the whole selection.

Catalogue

AimGel

	Volume	Particles	Cat. No.
Human T Cell Activator	0.5 mL	2×10^7	ATC01-0050H
Mouse T Cell Activator	0.5 mL	2×10^7	ATC01-0050M
Dog T Cell Activator	0.5 mL	2×10^7	ATC01-0050D
Human NK Cell Activator	0.5 mL	2×10^7	ANK01-0050H

AimCore

Degradability	Membrane fluidity	Avidin amount	Volume	Particles	Cat. No.
Hydrolyses in 7-10 days under physiological conditions	Low	~4,000/µm²	1.0 mL	4×10^7	AVI01-0100
		~14,500/µm²	1.0 mL	4×10^7	AVI02-0100
		~44,000/µm²	1.0 mL	4×10^7	AVI03-0100
	High	~4,000/µm²	1.0 mL	4×10^7	AVI04-0100
		~14,500/µm²	1.0 mL	4×10^7	AVI05-0100
		~44,000/µm²	1.0 mL	4×10^7	AVI06-0100
Enzyme lysable at any timepoint	Low	~4,000/µm²	1.0 mL	4×10^7	AVI07-0100
		~14,500/µm²	1.0 mL	4×10^7	AVI08-0100
		~44,000/µm²	1.0 mL	4×10^7	AVI09-0100
	High	~4,000/µm²	1.0 mL	4×10^7	AVI10-0100
		~14,500/µm²	1.0 mL	4×10^7	AVI11-0100
		~44,000/µm²	1.0 mL	4×10^7	AVI12-0100

Not seeing your cell type or species?

Or want AimGel activators that self-hydrolyse on a schedule matched to your culture conditions? Reach out to welovecells@allegrowbiotech.com for custom formulations.

All items ship on ice; contents sterile in unopened tube. Do not freeze. See individual protocols for handling limits.

See pricing at www.allegrowbiotech.com/webstore

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